Bubble detachment from circular cavities and flat surfaces

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Received 00th January 20xx, Accepted 00th January 20xx DOI: 10.1039/x0xx00000x

We determine the maximum stable volume of bubbles attached to flat surfaces as a function of contact radius, contact angle, and capillary length. By solving the Young–Laplace equation using a shooting method, we calculate equilibrium bubble shapes and determine when bubbles detach through three distinct modes: necking at the cavity, sideways instability, and spreading followed by necking on the surrounding surface. Our predicted detachment volumes agree well with experimental measurements of electrolytic, boiling, and carbonated bubbles, as well as pendant drops. These results bring together and extend previous analytical estimates and, by symmetry, also apply directly to pendant drops.

1 Introduction

1.1 Motivation

Annually, around 10^{11} kg of hydrogen gas is used in industry, mostly to produce fertilisers via the Haber process and to reduce metal ores in steelmaking and aluminium production [1]. Currently, more than 95% of hydrogen is produced from fossil fuels, largely from natural gas via steam methane reforming [2], which releases significant quantities of CO_2. Diversifying away from fossil fuels is essential to maintain a stable Earth climate system and ensure the long-term survival of human society [3]. Water electrolysis driven by renewable electricity is a promising alternative [1], splitting water into hydrogen and oxygen without direct CO_2 emissions. However, gas bubbles that form and remain attached to the electrode surface limit the efficiency of water electrolysis. These bubbles block the active area, impede ion transport, and increase ohmic resistance [4–8]. Electrolysis will also be important for life-support systems on future manned lunar and Martian bases, where water extracted from local ice deposits could provide a source of hydrogen and oxygen [9]. Understanding and controlling bubble detachment is therefore relevant both to improving electrolysis efficiency on Earth and to enabling efficient resource utilisation in low-gravity environments [10].

Bubble detachment also plays a role in the transport of naturally occurring gases. Methane seepage from the seafloor releases an estimated 6–12 Mt of methane per year from anoxic sediments, geological reservoirs, and degrading methane-hydrate deposits [11]. Bubbles with radii below approximately 3 mm can dissolve and biodegrade in the water column before reaching the ocean surface [12], whereas larger bubbles can escape into the atmosphere. Since methane is a potent greenhouse gas, understanding the conditions governing bubble detachment and subsequent transport is relevant to quantifying its environmental impact.

1.2 Background

For a bubble attached to a horizontal surface (illustrated in Figure 1) the scale over which surface tension σ balances fluid density contrast $\rho_{\text{out}} - \rho_{\text{in}}$ and gravity g is the capillary length [13],

$$\lambda \equiv \sqrt{\frac{\sigma}{(\rho_{\text{out}} - \rho_{\text{in}})g}}. \tag{1}$$

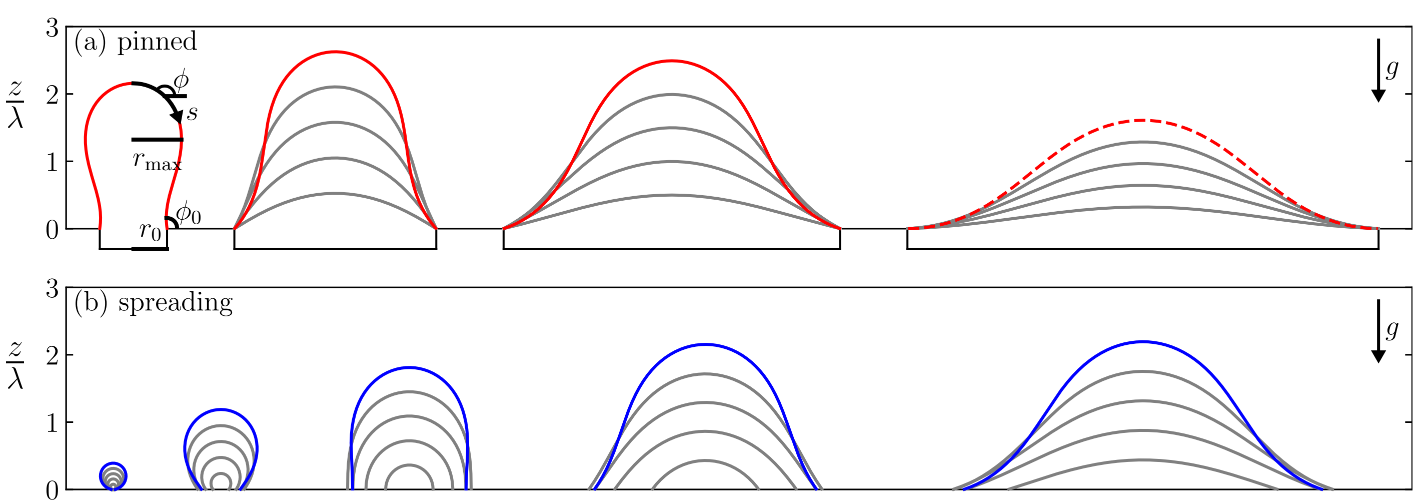

Figure 1 (a) Profiles of pinned bubbles with contact radius $r_0/\lambda \in 0.5, 1.5, 2.5, 3.5$, from left to right. The largest pinned bubble for each contact radius is shown in red. The $r_0 = 3.5\lambda$ profile is dashed to indicate the limiting case $\phi_0 = \pi$, for which the bubble detaches via a non-axisymmetric instability. The coordinate system (ϕ, s) used to solve Equation 11 and the bubble properties ϕ_0, r_0, and $r_{\max}$ are shown in black. (b) Profiles of spreading bubbles with contact angle $\phi_0/\pi \in 0.1, 0.3, 0.5, 0.9$, from left to right. The largest spreading bubble for each contact angle is shown in blue.

The capillary length sets the radius of curvature of the bubble interface due to gravity. Two distinct attachment modes are recognised in the literature[14]. *Pinned* bubbles have a fixed contact radius r_0, with contact angle ϕ_0 changing as the bubble inflates. *Spreading* bubbles have a fixed contact angle ϕ_0, with contact radius r_0 changing as the bubble inflates.
For *pinned* bubbles, the classical Tate volume[15] assumes that a bubble detaches when the contact angle reaches $\phi_0 = \pi/2$, balancing buoyancy and surface tension to give

$$V = 2\pi\lambda^2 r_0. \tag{2}$$

Chesters[16] refined this in the limit $r_0/\lambda \to 0$, also assuming detachment at $\phi_0 = \pi/2$. Subsequent experiments, however, show systematic deviations from both models[17–21], due to non-spherical interfaces and the onset of asymmetrical instabilities. Stability analyses indicate that the bubble becomes unstable to asymmetrical perturbations for cavity radii $r_0 > 3.219\lambda$[22], and no stable bubble exists at fixed volume for $r_0 > 3.832\lambda$[23]. For such large cavities, lateral detachment from the rim is observed experimentally[20].
The study of *spreading* bubble departure dates back to Fritz[24], who numerically estimated the maximum bubble volume as

$$V = 0.887\lambda^3\phi_0^3 \tag{3}$$

in the limit of small contact angles ϕ_0. Focusing instead on large contact angles, Phan et al.[25] developed an empirical model for bubble departure from nanocoated surfaces. More recent models account for contact-angle hysteresis[26], using advancing, receding[27,28], or dynamic[29] angles as inputs. Experiments, however, reveal systematic deviations from these predictions, particularly for bubbles on superhydrophobic and high-contact-angle surfaces[30].
Bubble detachment can also be triggered by coalescence with a neighbouring bubble. Whether neighbouring bubbles coalesce is determined by their width relative to the distance between nucleation sites[31,37]. Following coalescence, the merged bubble may either remain attached or detach from the surface, depending on the balance between adhesion and viscous dissipation[32].

1.3 Aims

Here, we treat pinned and spreading bubbles together, rather than as separate problems. We address three fundamental questions:
Q1 What is the volume of the bubble when it detaches?
Q2 How wide does the bubble grow before it detaches?
Q3 How does it detach?
We consider a simplified system consisting of a circular cavity of radius r_0 containing an already-nucleated bubble, surrounded by a smooth horizontal surface with a single-valued, non-hysteretic contact angle ϕ_0. That is, the contact angle is independent of whether the three-phase contact line advances or recedes. The fluid is quiescent, so forces are purely hydrostatic, and the system is physically equivalent to a pendant drop. Throughout this paper, we refer to it as a bubble, but our results apply equally to pendant drops. We determine the detachment size for pinned and spreading bubbles across the full range of cavity sizes, capillary lengths ($0 < r_0/\lambda < \infty$), and contact angles ($0 < \phi_0 < \pi$).

2 Methods

2.1 Governing equation

To calculate the bubble shape, we use and extend the method developed by Fritz[24]. Consider a bubble attached to a solid horizontal surface, as depicted in Figure 1. The bubble interface satisfies the Young–Laplace equation[33]:

$$\sigma\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = p_{\text{in}} - p_{\text{out}}, \tag{4}$$

where R_1 and R_2 are the principal radii of curvature at a point on the interface, and p_{in} and p_{out} are the pressures inside and outside the bubble, respectively. Assuming both fluid phases are in hydrostatic equilibrium, the pressures at height z are given in terms of the reference pressures $p_{0\text{in}}$ and $p_{0\text{out}}$ at the substrate ($z = 0$):

$$p_{\text{in}}(z) = p_{\text{in}0} - \rho_{\text{in}} g z, \tag{5}$$
$$p_{\text{out}}(z) = p_{\text{out}0} - \rho_{\text{out}} g z. \tag{6}$$

Substituting into Equation 4 and dividing through by the surface tension gives:

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{p_{\text{in}0} - p_{\text{out}0}}{\sigma} + \frac{z}{\lambda^2}. \tag{7}$$

To eliminate the pressure difference $p_{\text{in}0} - p_{\text{out}0}$, we evaluate Equation 7 at the apex of the bubble, where $z = h$ and the interface is spherical, so we let $R_1 = R_2 = R_h$, the apex radius of curvature:

$$\frac{2}{R_h} = \frac{p_{\text{in}0} - p_{\text{out}0}}{\sigma} + \frac{h}{\lambda^2}. \tag{8}$$

Subtracting this from Equation 7 gives:

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{2}{R_h} + \frac{z-h}{\lambda^2}. \tag{9}$$

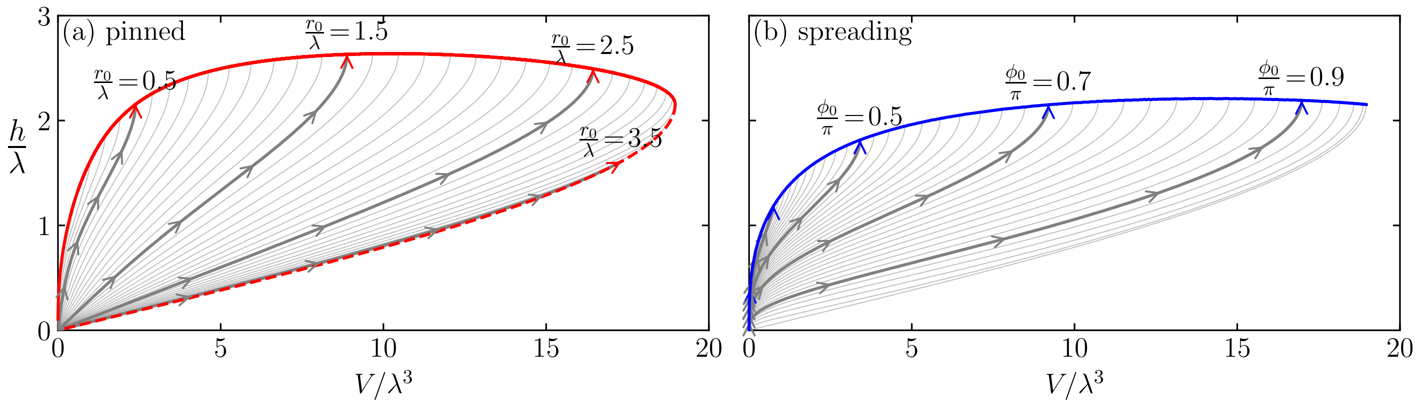

Figure 2 Bubble height h and volume V during evolution. (a) Grey lines: pinned bubbles with constant contact radius r_0. Solid red line: locus where $\partial V/\partial h=0$, and the bubbles detach axisymmetrically. Dashed red line: locus where $\phi_0=\pi$ and the bubbles detach from the side of the cavity. (b) Grey lines: spreading bubbles with constant contact angle ϕ_0. Solid blue line: locus where $\partial V/\partial h=0$, and the bubbles detach axisymmetrically. In (a) and (b), thick grey lines and arrows correspond to the bubbles shown in Figure 1.

Assuming the bubble is axisymmetric about the vertical axis, we describe the interface shape in terms of arc length s from the apex, distance from the axis r, and the angle ϕ between the interface tangent and the horizontal. The two principal radii of curvature can be found through trigonometry[34]:

$$R_1=\frac{ds}{d\phi}, \qquad R_2=\frac{r}{\sin\phi}. \tag{10}$$

Substituting into Equation 9 and multiplying by the capillary length yields the governing equation for the bubble shape:

$$\lambda\frac{d\phi}{ds}+\frac{\lambda\sin\phi}{r}=\frac{2\lambda}{R_h}+\frac{z-h}{\lambda}. \tag{11}$$

This equation describes the curvature of an axisymmetric interface under gravity. Here, all lengths are scaled by the capillary length λ, so that Equation 11 is dimensionless, implying self-similar bubble shapes and a universal scaling with λ.

2.2 Numerical solution

We solve Equation 11 using a shooting method, initiating the integration at the apex ($r=0$, $z=h$, $\phi=\pi$) and sampling 10^4 logarithmically spaced initial guesses for R_h over $10^{-3}\lambda<R_h<10^3\lambda$. The equations are integrated along the arc length s with a step size of $10^{-5}\lambda$ using the Adams–Bashforth method[34]. The complete set of calculations requires approximately 250 s on a single core of an M4 MacBook Pro. The C code and accompanying data are freely available[35].
For pinned bubbles, the boundary condition $r=r_0$ is imposed at the substrate ($z=0$). For a given cavity radius r_0, this yields a family of equilibrium profiles with different apex radii R_h, shown in Figure 1a. For spreading bubbles, the boundary condition $\phi=\phi_0$ is imposed at the substrate ($z=0$). The resulting profiles are shown in Figure 1b for a sequence of contact angles and apex radii R_h.

3 Results and Discussion

3.1 Bubble detachment criteria

To characterise the evolution of a growing pinned bubble, we plot the bubble height h as a function of volume V in Figure 2a, where the bubble volume

$$V=\int_0^h \pi r^2\,\mathrm{d}z, \tag{12}$$

is integrated using the midpoint rule. Each family of solutions in Figure 1a gives rise to a curve $h(V)$ in Figure 2a. Upon nucleation inside the cavity, the bubble has zero volume and zero height, corresponding to the bottom-left point in Figure 2a. As the bubble grows, its volume V and height h increase, and it remains attached to the substrate until one of two conditions is met:

1. The bubble volume reaches a point where the $h(V)$ curve becomes vertical (i.e., $\partial V/\partial h=0$), meaning the height can increase without a further increase in volume. The solution is no longer unique, and the interface becomes unstable, leading to axisymmetric detachment.

2. The contact angle ϕ_0 reaches π, at which point the interface is horizontal at the contact line and becomes unstable to non-axisymmetric perturbations[22]. The bubble shifts to one side of the cavity and detaches laterally. This occurs for all $h(V)$ curves with $r_0>3.219\lambda$, at the lower right of Figure 2a.

We truncate each of the $h(V)$ curves in Figure 2a when either of these conditions is met. We note that the tallest possible bubble occurs in a cavity of radius $r_0=1.720\lambda$ and has height $h=2.642\lambda$, which is considerably less than the analytical limit of $h=3.42\lambda$ established for two-dimensional columnar pendant drops[36].

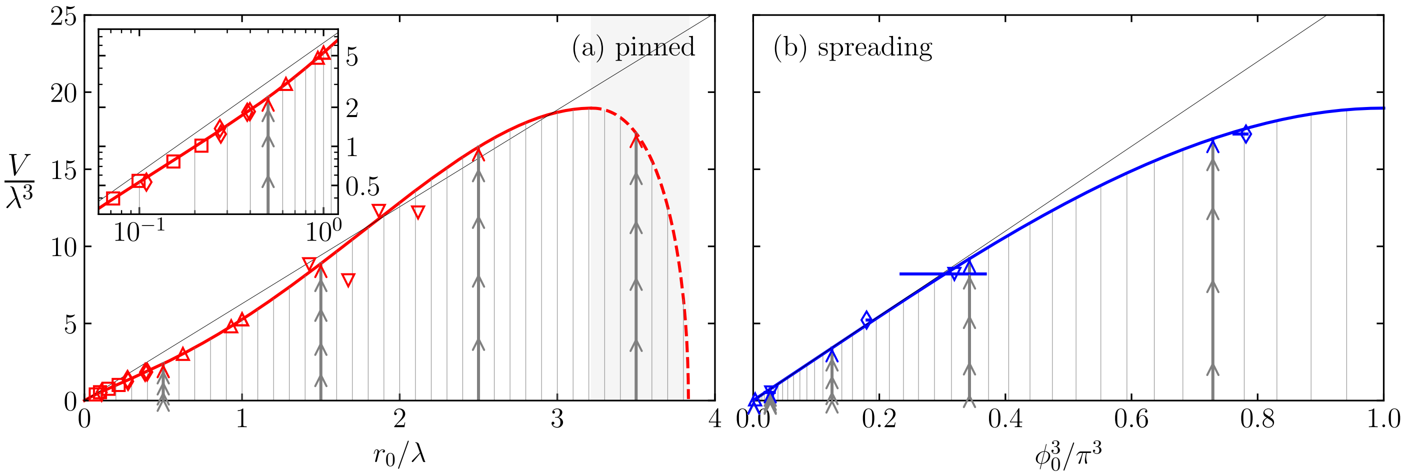

Figure 3 (a) Volume V of pinned bubbles. The red line shows the detachment volume as a function of contact radius r_0, while the thin black line, $V = 2\pi\lambda^2 r_0$, shows the Tate prediction[15]. The shaded region $3.219\lambda < r_0 < 3.832\lambda$ and dashed red line indicate where ϕ_0 reaches π, and the bubble becomes unstable to asymmetric perturbations. Empty symbols denote detachment volumes measured in experiments: squares[18], diamonds[19], down-triangles[20], and up-triangles[21]. Inset: same data on a logarithmic scale, showing the small-radius range in greater detail. (b) volume V of spreading bubbles. The solid blue line shows the detachment volume as a function of contact angle ϕ_0, which is cubed to allow comparison with the Fritz[24] prediction $V = 0.887\lambda^3\phi_0^3$ (thin black line). Empty symbols denote detachment volumes measured in experiments: up-triangles[28], down-triangles[27], and diamonds[30]; error bars indicate advancing and receding contact angles. In (a) and (b), thick grey lines and arrows correspond to the bubbles shown in Figure 1.

In Figure 2b, we perform the same $h(V)$ analysis for spreading bubbles. The figure shows the possible volumes and heights for contact angles spanning the full range $0 < \phi_0 < \pi$. As for pinned bubbles, each curve reaches a point where $h(V)$ becomes vertical ($\partial V/\partial h = 0$). This point defines the detachment volume, and the curve is truncated there. In the spreading bubble case, detachment is always axisymmetric, and the tallest spreading bubble is shorter, with $h = 2.213\lambda$ and contact angle $\phi_0 = 0.8330\pi$.

3.2 Bubble volume

We take the detachment volume to be the entire bubble volume above the contact line at the moment of detachment. We note, however, that some gas may remain attached to the surface after detachment, so the true released volume may be slightly smaller.

Figure 3a shows the detachment volume of pinned bubbles. It initially increases with contact radius, closely following the Tate prediction[15]; deviations from this prediction arise from the non-spherical interface shape induced by buoyancy. These deviations are consistent with experimental measurements of bubbles emitted from tubes in water[18], bubbles in carbonated water from millimetric cavities[19], and pendant drops[21]. The largest pinned bubble occurs at $r_0 = 3.219\lambda$, with a volume of $V = 18.96\lambda^3$. Beyond this point, for $r_0 > 3.219\lambda$, ϕ_0 reaches π and the interface becomes unstable to non-axisymmetric perturbations[22], causing the bubble to slide to one side and detach laterally; the detachment volume then decreases with increasing contact radius. No stable solution exists for $r_0 > 3.832\lambda$[23]: above this radius, Equation 11 has no solution with $\phi_0 < \pi$, and lateral detachment occurs immediately, consistent with the dripping behaviour reported in sessile drop experiments[20].

Figure 3b shows the detachment volumes of spreading bubbles. For small contact angles, the Fritz prediction is accurate. As $\phi_0 \to \pi$, however, the detachment volume saturates below Fritz's prediction. Our results are confirmed by measurements of water electrolysis[28], boiling[27], and air bubbles over superhydrophobic surfaces[30]. The largest spreading bubble has $V = 18.96\lambda^3$, and is identical in shape to the largest pinned bubble.

3.3 Bubble coalescence

When two bubbles nucleate near each other on a surface, they may coalesce, and the energy released during coalescence can lead to detachment[32]. Assuming that coalescence occurs when neighbouring bubbles touch, and that the bubbles have the same shape, coalescence occurs when their separation is less than or equal to $2r_{\max}$, where $r_{\max}$ is the maximum horizontal radius of the bubble[37], as labelled in Figure 1. Figure 4a shows the maximum value of $r_{\max}$ reached by pinned bubbles during their evolution up to detachment. The maximum horizontal radius follows the piecewise fit:

$$\frac{r_{\max}}{\lambda} \approx \begin{cases} (r_0/\lambda)^{1/3} & \text{for } r_0 < \lambda, \\ r_0/\lambda & \text{for } r_0 > \lambda. \end{cases} \tag{13}$$

For small contact radii, this means we expect bubbles to coalesce when the distance between their nucleation centres is less than $2r_0^{1/3}\lambda^{2/3}$. For large contact radii, $r_{\max} = r_0$ indicates that the bubble does not protrude beyond the cavity. Hence, coalescence cannot occur between bubbles nucleated on neighbouring cavities with radii greater than λ.

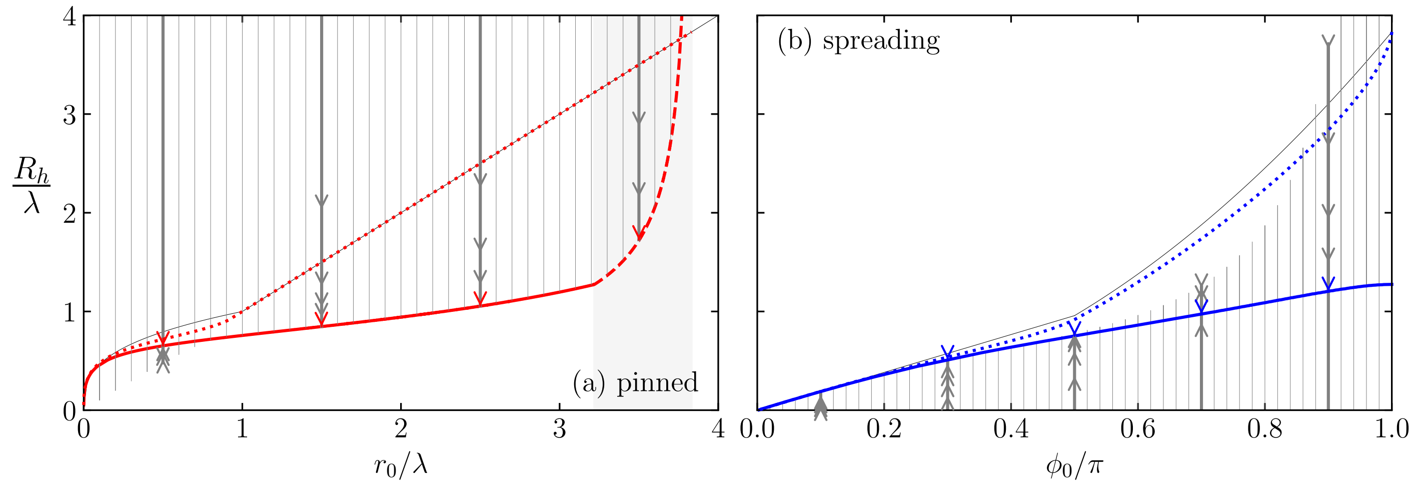

Figure 4 (a) Apex radius of curvature R_h of pinned bubbles as a function of contact radius r_0. The red line shows R_h at detachment. The shaded region $3.219\lambda < r_0 < 3.832\lambda$ and dashed red line indicate where ϕ_0 reaches π, and the bubble becomes unstable to asymmetric perturbations. The dotted red line shows the maximum horizontal radius $r_{\max}$ reached during evolution, and the thin black line shows the piecewise fit $r_{\max}/\lambda = (r_0/\lambda)^{1/3}$ for $r_0 < \lambda$ and $r_{\max} = r_0$ for $r_0 > \lambda$. (b) Apex radius of curvature R_h of spreading bubbles as a function of contact angle ϕ_0. The solid blue line shows R_h at detachment; the dotted blue line shows the maximum horizontal radius $r_{\max}$ reached during evolution; and the thin black line shows the piecewise fit $r_{\max}/\lambda = 3.832\phi_0/(2\pi)$ for $\phi_0 < \pi/2$ and $r_{\max}/\lambda = 3.832(\phi_0/\pi)^2$ for $\phi_0 > \pi/2$. In (a) and (b), vertical grey solid lines show the evolution of R_h up to detachment, with arrowheads corresponding to the bubbles shown in Figure 1.

The piecewise fit in Figure 4b shows that coalescence occurs when the separation d between nucleation sites is less than:

$$d < \begin{cases} 3.832\,\lambda\,\phi_0/\pi & \text{for } \phi_0 < \pi/2, \\ 2\times 3.832\,\lambda\,(\phi_0/\pi)^2 & \text{for } \phi_0 > \pi/2. \end{cases} \tag{14}$$

3.4 Bubble nucleation

Figure 4 also shows the evolution of the bubble apex radius of curvature R_h. Pinned bubbles nucleate with an infinite radius of curvature, corresponding to a flat interface spanning the cavity and, as follows from the Young–Laplace equation (4), zero pressure difference across the interface. In contrast, spreading bubbles nucleate with $R_h = 0$, corresponding to an infinitesimal spherical cap and a diverging pressure difference. This large pressure difference helps explain why nucleation of spreading bubbles is energetically unfavourable. Instead, nucleation generally occurs as a pinned bubble within a microscopic cavity.

3.5 Unpinning of the contact line

Figure 5a tracks how the contact angle of pinned bubbles evolves up to detachment. Every pinned bubble starts as a flat interface, so $\phi_0 = \pi$ initially. As the bubble inflates, the contact angle decreases until it reaches a minimum, coinciding with the appearance of a buoyancy-induced inflection point near the bubble's foot (visible in Figure 1), which then causes the contact angle to increase again before detachment. When the contact radius is small ($r_0 < 0.1\lambda$), the contact angle at detachment is close to $\pi/2$, consistent with the assumptions made by Tate and Chesters[15,16]. For larger contact radii, however, ϕ_0 at detachment exceeds $\pi/2$. This is why the Tate and Chesters predictions agree well with our results and experiments of others[18–21] for small contact radii but diverge as buoyancy effects grow stronger.

Figure 5b tracks how the contact radius r_0 of spreading bubbles evolves up to detachment. These bubbles start with $r_0 = 0$, and the contact radius grows as they inflate. As with pinned bubbles, buoyancy induces an inflection point near the bubble's foot, causing the contact radius to reach a maximum before the contact line recedes and the bubble detaches. In the (r_0, ϕ_0) plane, the locus of maximum contact radii for spreading bubbles coincides with the locus of minimum contact angles for pinned bubbles, and both are well described by the fit $\phi_0/\pi \approx \sqrt{r_0/(3.5\lambda)}$. At detachment, the contact radius scales roughly as the square of the contact angle, following $r_0 = 3.219\lambda(\phi_0/\pi)^2$.

Up to this point in the discussion, we have considered the results for pinned and spreading bubbles separately. Bubbles typically nucleate in the pinned mode[38] with $r_0 = r_c$, the radius of the cavity. However, the contact line will become unpinned if the contact angle ϕ_0 reaches the wetting angle ϕ_w of the three-phase system[14], it will then spread with constant contact angle $\phi_0 = \phi_w$. An example case $r_c = 0.5\lambda, \phi_w = 0.9\pi$ is illustrated in Figure 5b; the bubble nucleates pinned, and during evolution it becomes unpinned, and its contact line spreads across the surface surrounding the cavity before it reaches the maximum radius, recedes a little, and finally detaches. In general, we can see that the unpinning transition will occur if the crossover between the pinned mode $r_0 = r_c$ and the spreading mode $\phi_0 = \phi_w$ is reached before the minimum contact angle is reached, i.e., if the crossover point is above the line of best fit:

$$r_c < 3.5\,\lambda\left(\frac{\phi_w}{\pi}\right)^2. \tag{15}$$

Otherwise, the wetting angle will not be reached, and the bubble will remain in the pinned mode until detachment. Detachment from the pinned mode occurs axisymmetrically if the cavity radius $r_c < 3.219\lambda$, and non-axisymmetrically for large cavities $r_c \geq 3.219\lambda$. Figure 6

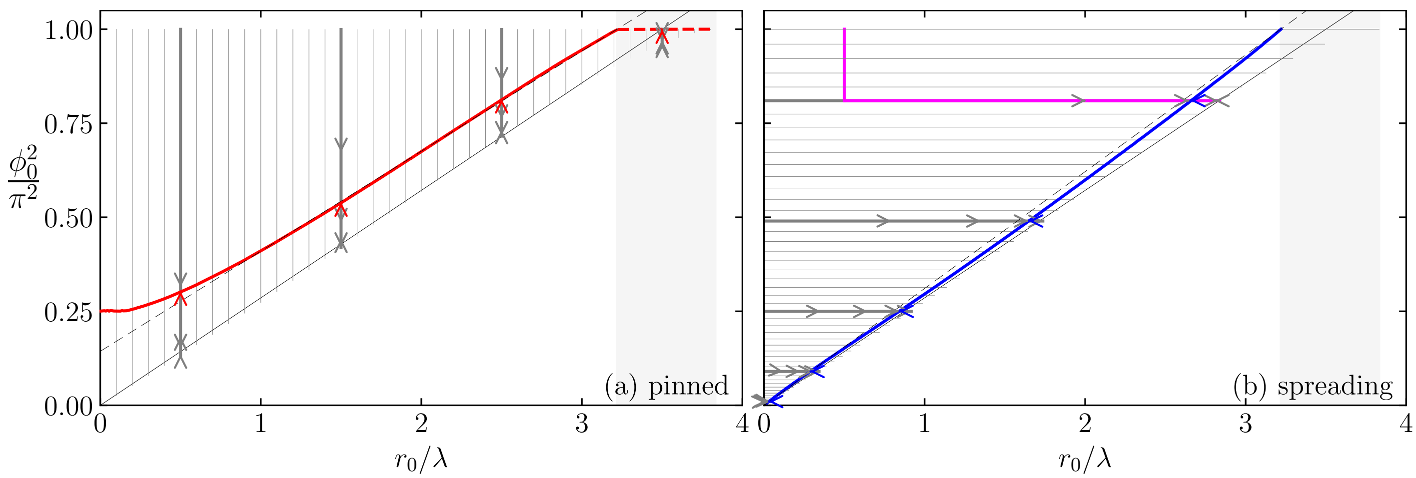

Figure 5 (a) Contact angle ϕ_0 of pinned bubbles with fixed contact radius r_0. The red line shows ϕ_0 at detachment and is dashed where $\phi_0 = \pi$ and detachment occurs non-axisymmetrically. The dashed black line shows the fit $(\phi_0/\pi)^2 = 0.2661 r_0/\lambda + 0.1436$. (b) Contact radius r_0 of spreading bubbles with fixed contact angle ϕ_0. The magenta line is the complete bubble evolution in a system with cavity radius $r_c = 0.5\lambda$ and wetting angle $\phi_w = 0.9\pi$. The blue line shows r_0 at detachment, with the dashed black line showing the fit $r_0 = 3.219\lambda(\phi_0/\pi)^2$. In (a) and (b), grey lines and arrows show the evolution up to detachment, the thin solid line $r_0 = 3.5\lambda(\phi_0/\pi)^2$ is a fit to the turning points, and the shaded region indicates $3.219\lambda < r_0 < 3.832\lambda$.

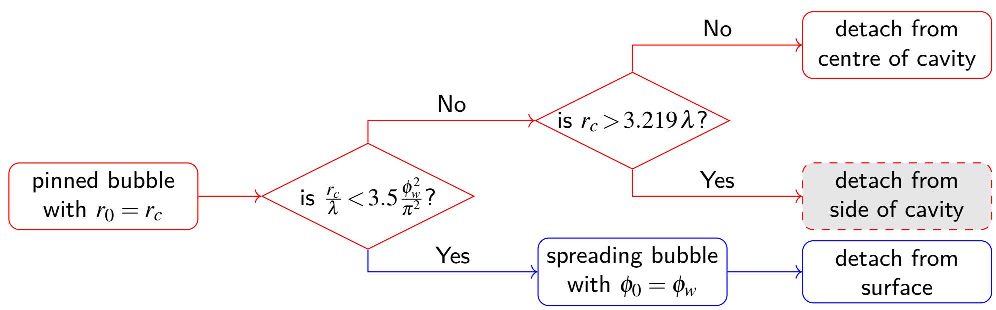

Figure 6 Decision tree for bubble detachment. Red branches correspond to pinned bubbles and blue branches to spreading bubbles. The cavity radius is r_c, and the wetting angle on the surrounding substrate is ϕ_w.

summarises these results in a decision tree for bubble evolution up to detachment. Given ϕ_0, r_0, and the capillary length λ of a specific system, the detachment size is determined by following this tree, using the functions plotted in Figure 3. Table 1 gives the properties of the limiting bubble configurations, where we see that the largest pinned and spreading bubbles in fact correspond to the same solution of Equation 11.

4 Conclusions

4.1 Answers to the three key questions

We now return to the three questions posed at the outset.

Answer to Q1. The detachment volume depends on whether the bubble is pinned or spreading. For pinned bubbles, the detachment volume rises nearly linearly with cavity radius, closely following the Tate prediction $V = 2\pi\lambda^2 r_0$ up to $V = 18.96\lambda^3$ with quantitative corrections arising from the non-spherical interface shape under buoyancy (Figure 3a). For cavities with $3.219 < r_0/\lambda < 3.832$, the detachment volume is $V < 18.96\lambda^3$, and for cavities larger than 3.832λ the detachment volume is zero as no stable bubble can exist. For spreading bubbles, the detachment volume increases monotonically with contact angle and saturates at $V = 18.96\lambda^3$ as $\phi_0 \to \pi$, below the Fritz prediction

Table 1 Key properties of significant bubbles. Shown are the contact angle ϕ_0, contact radius r_0, apex height h, apex radius of curvature R_h, and volume V, expressed in units of the capillary length λ. The corresponding radius r_{det} of the detached bubble is given for water at standard temperature and pressure, where $\lambda = 2.793$ mm. Extremal values are typeset in boldface.

Significance	ϕ_0/π	r_0/λ	h/λ	R_h/λ	V/λ^3	r_{det}/mm
largest pinned bubble	1.000	3.219	2.152	1.274	**18.96**	**4.620**
largest spreading bubble	1.000	3.219	2.152	1.274	**18.96**	**4.620**
tallest pinned bubble	0.7750	1.720	**2.642**	0.8867	10.62	3.808
tallest spreading bubble	0.8330	2.297	**2.213**	1.126	14.48	4.223
largest stable contact radius	1.000	**3.832**	0.000	∞	0.000	0.000

Table 2 Bubble detachment modes based on cavity radius r_c, wetting angle ϕ_w, and capillary length λ.

Cavity radius	Bubble mode	Detachment mode
$r_c < 3.5\,\lambda\,(\phi_w/\pi)^2$	spreading	straight upwards
$3.5\,\lambda\,(\phi_w/\pi)^2 \le r_c < 3.219\,\lambda$	pinned	straight upwards
$3.219\,\lambda \le r_c < 3.832\,\lambda$	pinned	sideways
$r_c \ge 3.832\,\lambda$	no stable bubble	sideways

(Figure 3b). Both the pinned and spreading results are confirmed by independent experimental datasets spanning electrolytic, boiling, and carbonated bubbles, as well as pendant drops.

Answer to Q2. For pinned bubbles with small contact radii ($r_0 < \lambda$), the width of the bubble is $\approx 2r_0^{1/3}\lambda^{2/3}$. However, for larger contact radii, the bubble is no larger than its contact patch. For spreading bubbles, the width closely approximates the piecewise function $2\times 3.832\,\lambda\phi_0/(2\pi)$ for $\phi_0 < \pi/2$ and $2\times 3.832\,\lambda(\phi_0/\pi)^2$ for $\phi_0 > \pi/2$. These empirical findings can be used to control and predict bubble coalescence.

Answer to Q3. The detachment mode is determined by which threshold is reached first during bubble growth (Figure 6). The behaviour depends strictly on the cavity radius r_c relative to the wetting angle ϕ_w and the capillary length λ, as summarised in Table 2.

4.2 Outlook

In summary, we have computed the maximum stable volume of axisymmetric bubbles on flat surfaces as a function of contact radius and contact angle, using a numerical solution of the Young–Laplace equation. The results unify the pinned and spreading detachment modes within a single framework, recover the classical Tate and Fritz limits, and identify the thresholds at which non-axisymmetric instabilities govern departure. All results scale universally with the capillary length λ and apply equally to pendant drops. The numerical code used in this study is freely available[35]. It can provide an exact detachment criterion for full electrolyser simulations, such as that of Ross *et al.*[8], as well as quantitative guidance for designing electrodes with optimised wettability and cavity geometry to promote early bubble detachment and improve electrolysis efficiency[5,37]. In future work, the model can be extended to account for heterogeneous surfaces, which give rise to contact angle hysteresis and repinning of the contact line.

5 Data availability

The data and code that support the findings of this article are openly available on Zenodo at `https://doi.org/10.5281/zenodo.20720640` (ref. 35).

6 Author Contributions

I. Cannon: conceptualisation, software, formal analysis, visualisation, writing – original draft. S. Endres: software, writing – review & editing. M. Avila: conceptualisation, methodology, writing – review & editing. L. Mädler: conceptualisation, methodology, writing – review & editing.

7 Conflicts of interest

There are no conflicts to declare.

8 Acknowledgements

This research was supported by the University of Bremen and the State of Bremen within the Humans on Mars Initiative and partially supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy EXC-3036 The Martian Mindset, project number: 533607631. I. Cannon gratefully acknowledges financial support from the Alexander von Humboldt Foundation through a Humboldt Research Fellowship for Postdoctoral Researchers.

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